

MGT 458: Experiments in Organizations

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DESCRIPTION

Businesses are increasingly using experimentation in a wide range of areas including product pricing, marketing strategies, employee incentives and shaping consumer behavior. These experiments can be particularly effective if the interventions tested incorporate insights from behavioral economics. However, many companies currently lack the knowledge necessary to take advantage of experimentation. An understanding of how to design and run a good experiment will position students at the cutting edge of decision-making in business.

The goals of the course are for students to understand: (1) the role of experiments in business (2) how insights from behavioral economics can be incorporated into experimental interventions and (3) how to design and run an experiment in an organization.

We will begin by discussing the role of experiments in organizations and examining some recent examples. Each student will be responsible for presenting a paper that describes an experiment run in a company or non-profit organization.

Students will then work individually or in teams to develop an experimental question and design a randomized experiment to test their question of interest. In the design phase, we will discuss methods of experimental design as well as strategies for working with an organization to implement experiments successfully. Students will run pilot studies to inform their designs. At the end of the course, students will present a proposed experimental design and implementation strategy.

About half of class time will consist of lecture. The other half will be spent workshopping concepts from the lecture and developing students' proposed experiments. The course is designed to give students the practical tools they need to successfully run experiments in organizations.

OBJECTIVES

At the close of this course you will be able to:

- Understand the role of experimentation in organizations
- Incorporate insights from behavioral economics into experimental interventions
- Design and implement randomized experiments
- Develop experiments that address the interests and concerns of companies and non-profits

MATERIALS

Nudge, Richard H. Thaler and Cass R. Sunstein (ISBN 9780143115267)

The Why Axis, Uri Gneezy and John List (ISBN 1610393112)

Mixed signals: How Incentives Really Work, Uri Gneezy (ISBN ISBN 9780300255539)

Running Randomized Evaluations: A Practical Guide, Rachel Glennerster and Kudzai Takavarasha (ISBN 9780691159270), referred to below as GT

Additional papers used in the course will be made available electronically.

CLASS STRUCTURE: For most modules, you will first complete the assigned lectures and reading and then submit a response survey that builds on the assigned content. Our in-person class time will focus on synthesizing content from the modules and workshopping experimental ideas

ASSIGNMENTS

Response surveys and activities: Response surveys and activities based on the lecture and reading are due:

- **Mondays** at 11:59 pm PT for online modules
- **Fridays** at 11:59 pm PT for in person modules
- The goal of the surveys is to track understanding and generate discussion. There is not necessarily a “right answer”.

Article summary and analysis: Students will present a paper describing an experiment run in an organization. The reading list includes example papers. You will prepare presentations individually or with a partner. You will also provide feedback to presenters. You will provide feedback individually.

Experimental proposal: Over the course of the class, you will work individually or in a group of to develop a proposal for an experiment to be run in an organization. In module 9, students will make a formal presentation of their experimental “pitch” as if presenting to the key decision maker in the organization. Your grade will be based both on your final proposal and your development of the proposal throughout the class. You will also provide peer feedback to the presentations of your classmates. In module 10, you will submit your finalized proposal incorporating feedback you received in your presentation.

Pilot study: Over the course of the class, students will design, run and analyze a pilot study to provide initial feedback for their experimental proposals. In module 9, you will present the results of your pilot study. You will also provide feedback to other groups’ pilot studies. You will work in your experimental proposal group for the pilot study. In module 10, you will submit your finalized pilot study results incorporating feedback you received in your presentation.

GRADING

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| • In person participation and response surveys | 25% |
| • Article presentation and peer feedback | 25% |
| • Pilot study and peer feedback | 25% |
| • Final experimental proposal and peer feedback | 25% |

ACADEMIC INTEGRITY

Integrity of scholarship is essential for an academic community. As members of the Rady School, we pledge ourselves to uphold the highest ethical standards. The University expects that both faculty and students will honor this principle and in so doing protect the validity of University intellectual work. For students, this means that all academic work will be done by the individual to whom it is assigned, without unauthorized aid of any kind.

The complete UCSD Policy on Integrity of Scholarship can be viewed at: <http://senate.ucsd.edu/manual/appendices/appendix2.pdf>

STUDENTS WITH DISABILITIES

A student who has a disability or special need and requires an accommodation in order to have equal access to the classroom must register with the Office for Students with Disabilities (OSD). The OSD will determine what accommodations may be made and provide the necessary documentation to present to the faculty member.

The student must present the OSD letter of certification and OSD accommodation recommendation to the appropriate faculty member in order to initiate the request for accommodation in classes, examinations, or other academic program activities. **No accommodations can be implemented retroactively.**

Please visit the [OSD website](#) for further information or contact the Office for Students with Disabilities at (858) 534-4382 or fosorio@ucsd.edu.

COURSE SCHEDULE. Depending on the progress in class, some changes may occur during the quarter.

Module	Wk	Date	Meet*	Topics	Video	Readings	Assignment (Due Date)**
Module 0:	0	1/02-1/4	Online	- Introduction & experimental questions - Randomization and causal inference	- Course overview - Experimental questions - Randomization	“Act Like a Scientist” “Discipline of Business Experimentation”	- Before completing the response survey, complete the can you guess activity at the following link. Record your number correct (I): https://80000hours.org/articles/can-you-guess/ <i>Response survey:</i> Complete the survey after viewing the lecture and completing the reading. DUE 1/5
Module 1:	1	1/5 ~ 1/11	In-Person (1/11)	Behavioral economics & experimental design	Experimental design	<i>Nudge</i>, Introduction and Part I <i>The Why Axis</i>, Chapter 1 <i>Mixed Signals</i>, Introduction	<i>Response survey:</i> Complete the survey after viewing the lecture and completing the reading. DUE 1/9
Module 2:	2	1/12~1/18	Online	Reading experiments	No video	Disney reading: “Shared Social Responsibility: A Field Experiment in Pay-What-You-Want Pricing and Charitable Giving”	<i>Response survey:</i> Complete the survey after completing the reading. DUE 1/9 DELIVERABLE: Submit your first and second choice for the article you will summarize. Articles are assigned first come, first served. DUE 1/9
Module 3:	3	1/19~1/25	Online	Online & pilot studies	- Pilot studies - Intro to Qualtrics	“The surprising power of online experiments” “Online field experiments: A selective survey of methods”	<i>Response activity:</i> Complete the Qualtrics survey activity after viewing the lecture and completing the reading. DUE 1/26 DELIVERABLE: Submit your group members for the pilot study and final presentation. DUE 1/26

Module 4:	4	1/26~2/1	Online	Synthesizing experiments	No video	Your chosen article	<u>DELIVERABLE:</u> Submit your article video presentation following the “Guide for reading and presenting experiments.” DUE 2/2 <u>DELIVERABLE:</u> Sign up for the article you will peer review. DUE 2/2
Module 5:	5	2/2~2/8	In-Person (2/8)	- Ethics of experimentation - Pilot design workshop	Ethics of experimentation	“Facebook Under Fire” “OKCupid Plays with Love in User Experiments” - Were OkCupid's and Facebook's Experiments Unethical? - Professors' Research Project Stirs Political Outrage in Montana	<i>Response survey:</i> Complete the survey after viewing the lecture and completing the reading. DUE 2/5 <u>DELIVERABLE:</u> Submit your one-page pilot design (one per group) using the “Guide for pilot study designs”. DUE 2/5 <u>DELIVERABLE:</u> Bring 2 PRINTED COPIES of your pilot design to class. DUE 2/8
Module 6:	6	2/9~2/15	Online	Randomization design and assigning treatment	Assigning treatment	The article you are peer reviewing	<i>Response survey and activity:</i> Complete the survey and submit the assigning treatment activity after viewing the lecture. DUE 2/16 <u>DELIVERABLE:</u> Submit your article presentation peer review after viewing the video and reading the article you are peer reviewing. DUE 2/16
Module 7:	7	2/16~2/22	Online	Communicating experimental results	- Communicating experimental results - OPTIONAL: Sample sizes and power	Advertising Experiments at RestaurantGrades	<i>Response survey and activity:</i> Submit data results activity after viewing the lecture and completing the reading. DUE 2/23

Module 8:	8	2/23~3/1	Online	▪ Culture of experimentation	▪ Culture of experimentation	▪ No reading	▪ <i>Response survey:</i> Complete the survey after viewing the lecture. DUE 3/2 ▪ <u>DELIVERABLE:</u> Submit your pilot study data analysis. DUE 3/2
Module 9:	9	3/2~3/8	In-Person (3/8)	▪ Pilot study presentations & experimental pitches	▪ No video	▪ No reading	▪ <u>DELIVERABLE:</u> Submit the slides used for your final presentation. DUE 3/7 ▪ <u>DELIVERABLE:</u> Final presentation in class. DUE 3/8 8am.
Module 10:	10	3/9~3/13		▪ Final results & experimental designs	▪ No video	▪ No reading	▪ <u>DELIVERABLE:</u> Submit revised final presentation. DUE 3/13

*In-person (Location: **TBD**) and Online (Canvas and Office Hours)

**All assignments/assessments are due by 11:59 p.m. Pacific Time on the due date unless otherwise noted.

Disclaimer: Please note that the specifics of this Course Syllabus are subject to change, and you will be responsible for abiding by any such changes. Your instructor will notify you of any changes.

Supplemental reading:

Module 0: Glennerster and Takavarasha (GT), *Running Randomized Evaluations*, Chapter 2, modules 2.1-2.3 (pp. 24-47). These modules discuss both experimental and non-experimental methods for measuring *causal impact*. At the end of the reading, you should understand what causal impact is and be able to discuss the advantages and disadvantages of the different methods.

Module 0: Glennerster and Takavarasha (GT), *Running Randomized Evaluations*, Chapter 4, module 4.2 (pp. 108-120). This module provides a critical overview of randomization. At the end of the reading, you should be able to apply the concepts in the reading to the evaluation and design of experiments; in particular, choosing a level of randomization.

Module 6: Glennerster and Takavarasha (GT), *Running Randomized Evaluations*, Chapter 4, module 4.1 (pp. 98-108), 4.4-4.5 (pp. 140-161). These modules discuss opportunities for randomization, randomization designs, and methods for randomizing participants. At the end of the reading, you should be able to apply the concepts in the reading and implement randomizations, including understanding the difference between a lottery design and an encouragement design, as well as the role of stratification (or blocking) in randomizations to achieve balance.

Module 7: Reading: Glennerster and Takavarasha (GT) Chapter 6, modules 6.1-6.5. These modules discuss statistical analysis and statistical power. At the end of the reading, you should be able to apply the concepts to choosing appropriate sample sizes for experiments.